

Name - B. Vishnu

Course Code - CSA 1712

Reg. No. 192512410

Course Name - Artificial Intelligence

Assessment - Col AT2

1) Problem Formulation:

* Variables: D_1, D_2, D_3

* Domain: $\{\text{Morning (M), Afternoon (A), Night (N)}\}$

* Constraints:

i) $D_1 \neq N$ (D_1 cannot work Night)

ii) $D_2 < D_3$ (D_2 works before D_3 in the timeline $M < A < N$).

iii) $\text{AllDiff}(D_1, D_2, D_3)$ [only one doctor per shift; exactly one shift per doctor].

iv) $D_3 \neq M$ (D_3 cannot work morning).

Backtracking Search Steps:

We start with an empty assignment $\{\}$ and assign shifts one by one, checking constraints at each step.

* Step 1: Select Variable D_1 .

- Valid domain based on constraint 1 ($D_1 \neq N$) is $\{M, A\}$.
- Action: Assign $D_1 = M$.
- Assignment which is in consistent: $\{D_1: M\}$

* Step 2: Select Variable D_2 .

- Valid domain based on Constraint 3 (alldiff) is $\{A, N\}$ because M is taken.
- Action: Assign $D_2 = A$.
- Current Assignment: $\{D_1: M, D_2: A\}$

* Step 3: Select Variable D_3 .

- Valid domain based on Constraint 3 (alldiff) is $\{N\}$.
- Action: Assign $D_3 = N$.
- Constraint Check: Does $D_2 < D_3$?
Yes, Afternoon $<$ Night [Constraint 2 is Satisfied].
Does $D_3 \neq M$? Yes, $N \neq M$ (Constraint 4 is satisfied).
- ~~Constraint~~ Current Assignment: $\{D_1: M, D_2: A, D_3: N\}$

Final Valid schedule:

- * Morning Shift: Doctor 1 (D_1)
- * Afternoon Shift: Doctor 2 (D_2)
- * Night Shift: Doctor 3 (D_3)

2) Hypothetical Scenario:

- Start Node (S): (0,0)
- Goal Node (G): (2,2)
- Obstacle : (1,1)
- Cost : $g(n) = 1$ per move
- Heuristic (Manhattan Distance): $h(n) = |x_n - x_{goal}| + |y_n - y_{goal}|$
- Algorithm : Greedy Best-First Search [evaluates nodes based solely on $h(n)$ to minimize priority queue cost]

Step-by-Step Node Expansion:

* Initialization:

- Calculate $h(S)$: $|0-2| + |0-2| = 4$
- Priority Queue (PQ): $[S(0,0): h=4]$

* Iteration 1:

- Pop : $[S(0,0)]$
- Expand available neighbours: Up to (0,1).
Right to (1,0). [Assuming Down/Left are out of bounds].

- Calculate h for neighbors:

$$(a) h(0,1) = |0-2| + |1-2| = 3 \quad (b) h(1,0) = |1-2| + |0-2| = 3$$

- PQ Updates [sorted by h]:

$$[Node(0,1): h=3, Node(1,0): h=3].$$

* Iteration 2:

- Pop: Node(0,1)
- Expand neighbors: Up to (0,2), Right to (1,1) - skip.
- Calculate h : $h(0,2) = |0-2| + |2-2| = 2$
- PQ Updates: $[Node(0,2): h=2, Node(1,0): h=3]$.

* Iteration 3:

- Pop: Node(0,2)
- Expand neighbors: Right to (1,2)
- Calculate h : $h(1,2) = |1-2| + |2-2| = 1$
- PQ Updates: $[Node(1,2): h=1, Node(1,0): h=3]$

* Iteration 4:

- Pop: Node(1,2)
- Expand neighbors: Right to (2,2)
- Calculate h : $h(2,2) = |2-2| + |2-2| = 0$ [Goal Reached]

Final Outcome:

- Optimal Path: $S(0,0) \rightarrow (0,1) \rightarrow (0,2) \rightarrow (1,2) \rightarrow G(2,2)$
- Total Cost : $4 \text{ moves} \times 1 = 4$.

3) The robot operates in a partially observable disaster environment and must safely reach a survivor while minimizing total cost.

(A) Constraint Analysis:

Constraint 1: Move only Up, Down, Left, Right.

- Diagonal movement is not allowed.
- The robot must plan routes using only four directions.

Constraint 2: Obstacles

- Blocked rooms cannot be entered.
- The robot must search for alternative paths.

Constraint 3: Limited Sensing

- The robot can observe only neighbouring cells.
- It cannot see the entire map initially.
- The map is updated as the robot explores.

Constraint 4: Movement Cost

- Normal move = 1
- Risky zone = $1 + 2 = 3$

The robot should avoid risky zones whenever possible.

Constraint 5: Online Search

- The robot learns about the environment during movement.
- It replans whenever new obstacles are discovered.

(B) Solution Approach:

Suitable Search Strategy:

Uniform Cost Search [UCS] with Online Search Agent.

Steps:

- i) Start at S.
- ii) Observe neighbouring cells.
- iii) Expand the lowest-cost path.
- iv) Ignore blocked cells.
- v) Update the internal map after every move.

- v) Recalculate the lowest-cost route if new obstacles are found.
- vi) Continue until the survivor (G) is reached.

Why UCS? $\rightarrow$ Uniform Cost Search always expands the node with the smallest accumulated cost, making it suitable when movement costs vary because of risky zones.

c) AI Concepts:

Intelligent Agent:

The rescue robot is an intelligent agent:

- * perceives the environment,
- * processes sensor information,
- * plans actions,
- * performs actions to achieve its goal.

Rationality:

A rational agent chooses the action that maximizes success while minimizing total travel cost and avoiding unnecessary risk.

Environment Type:

- partially observable
- Dynamic
- Sequential
- Stochastic
- Discrete
- Grid Environment.

① Justification:

Uniform Cost Search combined with an Online Search Agent is the most suitable approach because:

- * It always finds the least-cost path.
- * It handles different movements correctly.
- * It adapts to newly discovered obstacles.
- * It supports real-time decision making in unknown environments.
- * It improves safety by avoiding costly risky zones whenever possible.

Conclusion:

The rescue robot continuously senses its surroundings, updates its knowledge, and selects the safest and least-cost path to reach the survivor.